E-Cigarette and Healthy Habits Study

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Introduction

E-cigarettes entered the market and gained popularity in the United States beginning in 2006. They were marketed as a healthier alternative to traditional cigarettes as they did not contain tar or smoke, and only produced vapor. In addition, their ability to include a wide variety of flavors such as fruits and candies widened their appeal to a younger demographic. With organizations such as D.A.R.E, the World Health Organization and the Center for Disease Control and Prevention looking to optimize their marketing to dissuade non-smoking audiences from adapting this behavior, or for traditional smokers to think this is a completely safe alternative, our team was focused on building a series of predictive models to ascertain which societal members were more or less likely to use e-cigarettes.

More specifically, we focused on trying to determine if populations that practiced more healthy behaviors than not, such as maintaining a healthy BMI, avoiding binge drinking and having health insurance, would be less likely to smoke this cigarette alternative. If our series of supervised and non-supervised models could conclusively demonstrate this belief to be true, it would provide valuable information to these organizations on how they should adjust their marketing efforts to change behaviors on audiences that are taking worse care of their health, and therefore less likely to be motivated by health focused messaging.

In the following report we will thoroughly explain the data we chose, how we selected and utilized our variables, the models we selected, the reasoning behind them and most importantly, the quantifiable results these models produced.

Background & Question

Question: Are those with healthier lifestyles, defined as individuals who were classified as exhibiting healthy outcomes in a majority (>50%) of the health-related variables, less likely to use e-cigarettes than those who have less healthy lifestyles?

Hypothesis: We hypothesize a predictive model for e-cigarette usage can be built using health and lifestyle data because unhealthy habits frequently co-occur.

Prediction: We predict that those with healthier living habits may be less likely to use e-cigarettes due to their desire to live the healthiest lifestyle.

Methods

Initially our team was looking at three supervised and one unsupervised model, a Logistic Regression as a baseline, a Ridge, a Random Forest and a LDA in an unsupervised manner. With a focus on a binary outcome, we thought a Logistic Regression made the most sense, while allowing us to set a baseline for measurements to weigh more complicated models against. With a strong potential in non-linear relationships being present, we believed these four models would allow us the best chance to cover all of our bases and address a wide variety of accuracy based metrics to analyze our hypothesis.

After additional thought, variable selection and exploratory data analysis, we pivoted the above plan to choose an Elastic Net Logistic Regression as our baseline and used in both a supervised and unsupervised way. Due to the large number of variables, over 90 once one hot encoding was done, we knew an LDA was no longer an option and the Elastic Net would allow for potential dimension reduction.

Elastic Net Logistic Regressions also allow you to construct them as Ridge and Lasso models with a simple change, meaning this process simplified our demands while still keeping our results robust. We decided to stick with a Random Forest model as well.

In terms of models assumptions, for an elastic net we are assuming coefficients are non-zero so they can be shrunk, linearity of relationships, normality of residuals, equal variances, no high influence or leverage and that the data collected randomly and independently. For the Random Forest we are also assuming that the data was collected randomly and independently and that there is low or no multicollinearity. The Random Forest model is non-parametric, meaning there are no assumptions made about the underlying data distribution. Additionally, we are assuming our M.I.C.E imputation was done correctly as we did need to do this before one hot encoding due to very long run times of this particular line of code, as well as using it in the data portion set aside to test models, or we wouldn't have been able to complete the elastic net. Lastly, with any survey, there can be bias, even when random sampling was conducted, so this is an additional assumption we are making.

We tested for multicollinearity but need to retest this with one hot encoding done. Previous analysis only looked at numeric variables. We also found no zero correlations in our elastic net model. We aren't able to test directly if the data was collected randomly and independently, but due to the data being collected by the CDC, we do trust the source that gathered the material.

For all four of our models, cross validation was done with 5 folds being completed for the random forest and 3 being done for the elastic net due to the long duration it takes to run those models. This means our data was divided, with each split rotating as part of the training data two to four times, and each being the test set once. This method is used to reduce chances of overfitting on the training data. We then average the results of these splits afterwards.

Results

Elastic Net Logistic Regression Baseline:

The baseline model used for this analysis was an Elastic Net Logistic Regression. This model was trained using the imputed, scaled, and one-hot encoded data. Five variables were removed based on the results from the unsupervised Lasso that was run previously: 'Age_Category_Elderly', Age_Category_Young_Adult', 'Education_College_4_years_or_more_(college_graduate)', 'Has_Disability', and 'Marital_Status_Married'. This model was run with an L1 Ratio of 0.5. An L1 ratio of 1, or an L1 penalty, is used exclusively for a Lasso model, where certain variables are set completely to 0 if they are not contributing to the model's performance. An L1 ratio of 0, also known as an L2 penalty, is used exclusively for a Ridge model, where a certain amount of noise is added to each variable to increase the model's performance. Setting this value initially to 0.5 allows for the model to not act for Lasso or Ridge and instead provide more of a baseline value. Once this model is trained, this value can be altered so that Lasso and Ridge regressions can be tested. Finally, for the baseline model, 100 iterations were chosen to train the model.

Once the model was trained, key metrics were obtained, including a Confusion Matrix, Precision, Recall, F1-Score, and an AUC-ROC Score and plot. The Confusion Matrix can be seen below in Figure 1.

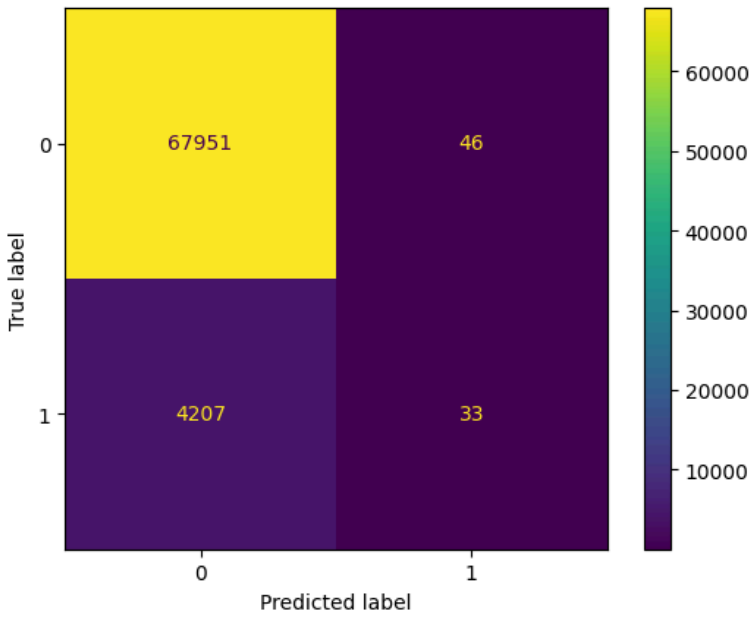

Figure 1: The Confusion Matrix for the baseline Elastic Net Logistic Regression.

The Confusion Matrix shown in Figure 1 shows that the model was able to correctly predict 67951 True Positive predictions, which would be 67951 predictions where the model was able to predict that an individual was an E-Cigarette user. It also was able to successfully predict 33 True Negatives, which would be 33 predictions where an individual was not an E-Cigarette user. The model, however, made 46 False Negative predictions, where 46 individuals were predicted to not be E-Cigarette users when they were actually E-Cigarette users. Finally, the model made 4207 False Positive predictions, meaning that 4207 users were incorrectly predicted as being an E-Cigarette user when they were actually not a user. As a result of this Confusion Matrix, Precision, Recall, F1-Score, and an AUC-ROC Score were calculated and can be seen in Figure 2.

Evaluation Metric	Value
Precision	0.42

Recall	0.01
F1-Score	0.02
AUC-ROC	0.78

Figure 2. Evaluation Metrics for the baseline Elastic Net Logistic Regression.

Figure 2 displays the evaluation metrics for the baseline Elastic Net Logistic Regression. It is important to note that positive results here indicate that a user is an E-Cigarette user. The Precision score is 0.42, which would indicate that less than half of the time, the model is correctly identifying positive results. A perfectly precise model would have a value of 1.0. The Recall score was 0.01, which indicates that only a very small portion of actual positive results were correctly classified as positive results. Similar to recall, a perfect value for recall would also be 1.0. Finally, F1-Score is the balance between recall and precision. In an ideal world, since a perfect recall and a perfect precision would be 1.0, this value would also be 1.0. In this case, however, the value is 0.01, which indicates that the model does not have a good balance between precision and recall.

Another metric displayed in Figure 2 is the AUC-ROC score. An AUC-ROC score represents the likelihood that if a random positive and a random negative class is provided to the model, the model will be able to select the positive class and rank it higher than the negative class. So, the AUC-ROC score indicates how likely the model is to make a True Positive prediction. The higher this value, the better the prediction. So, this model has an AUC-ROC score of 0.78, which indicates that the model will be able to correctly identify if an individual is

an E-cigarette user ~78% of the time. AUC-ROC can also be seen in the plot below in Figure 3.

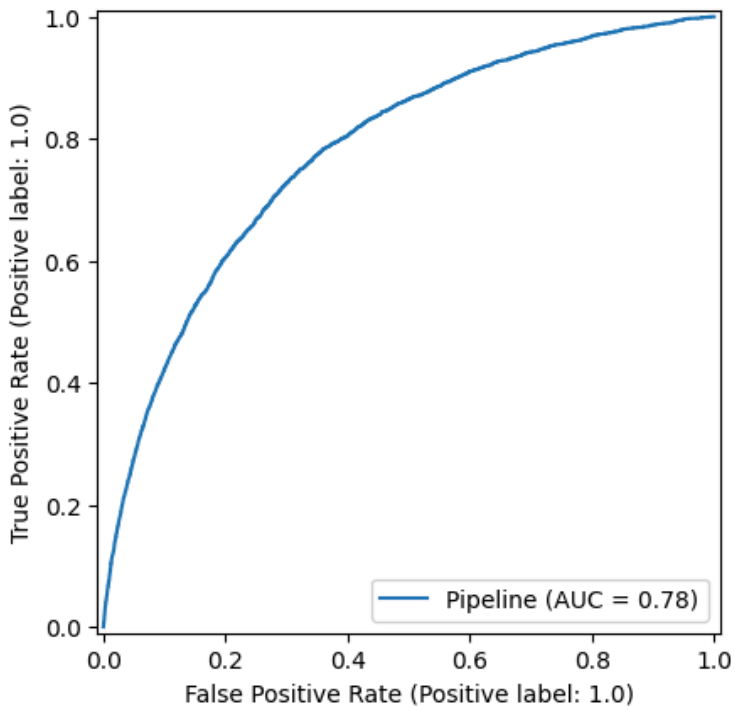

Figure 3. AUC-ROC curve for the baseline Elastic Net Logistic Regression.

Figure 3 displays the AUC-ROC curve for the baseline Elastic Net Logistic Regression. Ideally, a better model will have the AUC-ROC curve closer to 1, and the blue line on the plot will be much closer to the top left of the plot.

The baseline Elastic Net Logistic Regression displayed mixed results. Its AUC-ROC score was decent, but the precision, recall, and F1-scores were very poor. The model may be able to predict whether an individual is an E-cigarette user, but does not do a good job of predicting when an individual is not an E-cigarette user. There may be a few reasons for this. The first is that both the training and test sets were imputed. There were a lot of missing values in the dataset, so imputation was required to fill these values in. As a result, there may be additional bias that was introduced as a result of this imputation, which may be skewing the model's ability to make predictions and therefore lower its ability to make predictions for the negative class. The

other reason is that there may be a severe class imbalance between the positive class (being an E-cigarette user) and the negative class (not being an E-cigarette user). This would skew the model's ability to make predictions for each class equally. The results of the baseline Elastic Net Logistic Regression would indicate there is a clear imbalance, as the model can make predictions for the positive class but struggles to make predictions for the negative class. At this point, the penalty for the Elastic Net Logistic Regression was changed to be exclusively Ridge and exclusively Lasso to try to improve the results of the model and counteract these assumptions.

Elastic Net Logistic Regression Ridge/Lasso:

In order to run the Elastic Net Logistic Regression as a Ridge model or a Lasso model, the penalty had to be set to 0 for Ridge and 1 for Lasso. The Confusion Matrix, Evaluation Metrics, and AUC-ROC curve for the Ridge and Lasso models can be seen below in Figures 4, 5, and 6 respectively.

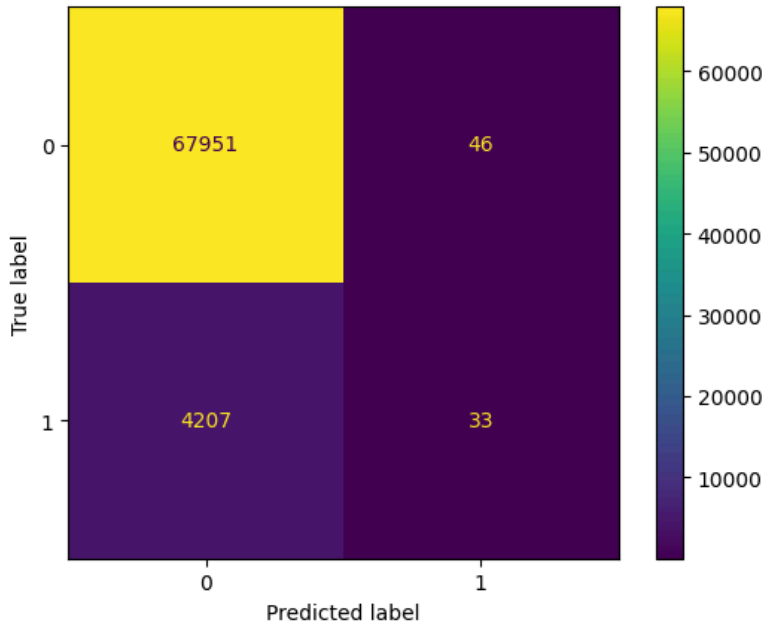

Figure 4. The Confusion Matrix for the Ridge and Lasso Elastic Net Logistic Regression.

Evaluation Metric	Value
Precision	0.42
Recall	0.01
F1-Score	0.02
AUC-ROC	0.78

Figure 5. Evaluation Metrics for the Ridge and Lasso Elastic Net Logistic Regression.

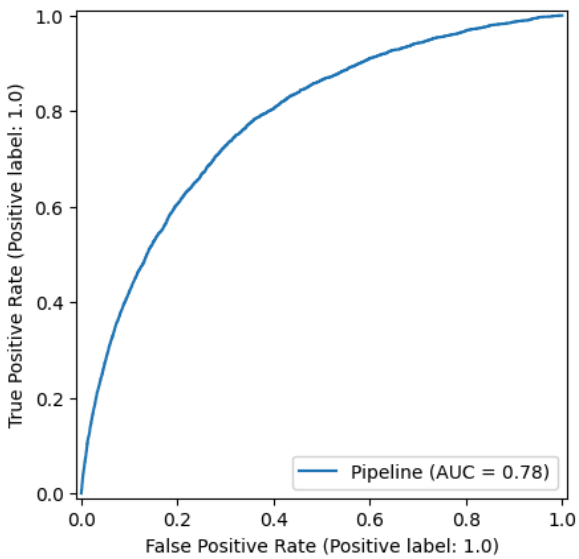

Figure 6. AUC-ROC curve for the Ridge and Lasso Elastic Net Logistic Regression.

Figures 4, 5, and 6 show the exact same results as Figures 1, 2 and 3. It is important to note that the results for Ridge and the Lasso were exactly the same, so the results were only included once. Altering the penalty score to account for Ridge and Lasso had no effect on the outcome of the model, which resulted in the exact same Confusion Matrix and therefore the same Evaluation Metrics. The AUC-ROC curve was exactly the same. These models may need to be tuned further. It is possible that there were not enough iterations built into the models initially to

allow them to converge on better values. Altering this hyperparameter will be one of the next steps to potentially improve the results of the model.

Random Forest:

A baseline Random Forest model was trained with 100 decision trees. From this model, evaluation metrics, including accuracy, precision, recall, and F1-score were produced. A confusion matrix and a ROC curve were also created, showing model performance. Lastly, feature importance was calculated, highlighting which variables contribute most to the model.

The random forest model was evaluated through k-fold cross-validation and used 5 folds. Since there is a heavy class imbalance and the target variable is binary, the F1-score was the evaluation metric used. The results can be seen in Figure 7.

Evaluation Metric	Value
Cross-Validation F1-Scores:	[0.001 0.002 0.001 0.0025 0.002]
Mean Cross-Validation F1-Score:	0.0017
Standard Deviation Cross-Validation F1-Score:	0.0006

Figure 7: Random Forest model cross-validation metrics for F1-Score.

The first calculation in the above table is the Cross-Validation F1-score for each of the 5 folds. The F1-score is the harmonic mean of the precision and recall and represents how well the model performs on a scale of 0 to 1 (*F1_score*). In this model, each of the five F1-scores are essentially zero. These results, combined with the small values for the mean and standard deviation F1-scores show that the baseline model has poor predictive performance. Cross-validation was also performed using recall as a metric, but those results were worse and can be found in the Appendix.

Accuracy, precision, recall, and F1-score were all calculated for the random forest model and the results can be seen in Figure 8. These results are directly impacted by the heavy class imbalance in our dataset.

Evaluation Metric	Value
Accuracy:	0.9413
Precision:	0.6
Recall:	0.0007
F1-Score:	0.0014

Figure 8: Evaluation metrics for the baseline Random Forest model.

Although the accuracy is very high, at 0.94, this is misleading due to the class imbalance. Once the class imbalance has been addressed, it is expected that the accuracy will change and potentially decrease. The precision is 0.6, meaning the model is correctly predicting the positive class 60% of the time. The recall is incredibly small, at 0.0007, indicating that the model does an extremely poor job at finding the positive class. Lastly, the F1-Score is 0.0014, indicating that the model, especially the precision and recall, are not performing well. Additional hyperparameter tuning along with handling of the class imbalance will help improve these results.

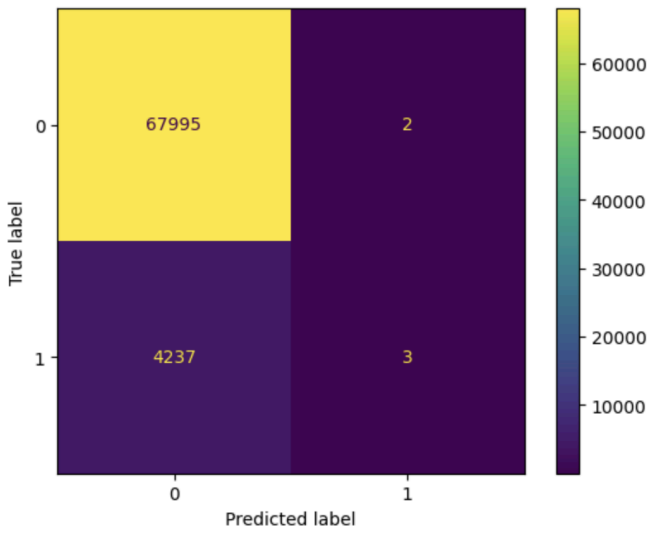

Figure 9: Confusion matrix for random forest model

The confusion matrix, as seen in Figure 9, shows the true and predicted labels of the target variable. The top left represents True Negative (TN), where the model correctly predicts non e-cigarette users. The top right represents False Positive (FP), where the model incorrectly predicts e-cigarette users, but it is actually non e-cigarette users. The bottom left represents False Negative (FN), where the model incorrectly predicts non e-cigarette users, but it is actually e-cigarette users. The bottom right represents True Positive (TP), where the model correctly predicts e-cigarette users.

The results of the confusion matrix further validate that our model performance is weak, is showing bias, and is almost always predicting non e-cigarette users. It only predicts e-cigarette users five times, which is incredibly low.

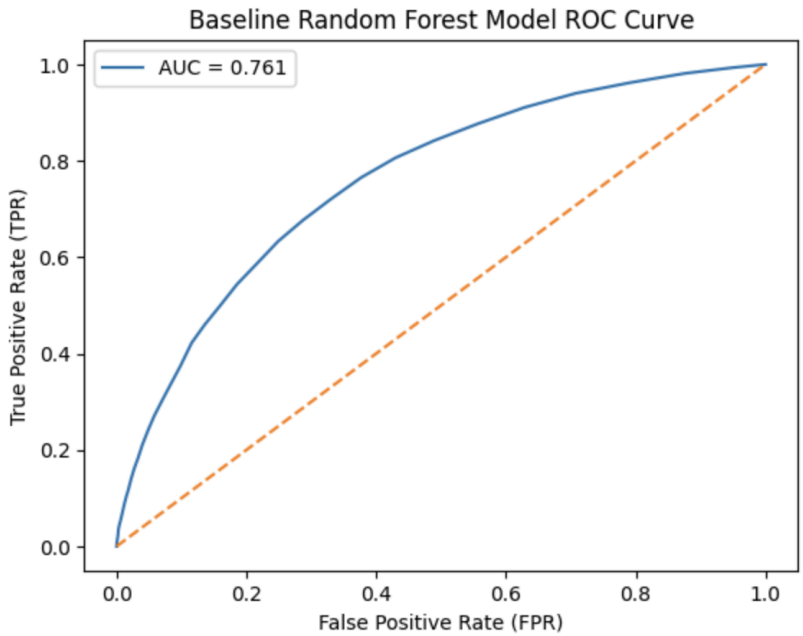

Figure 10: ROC Curve for random forest model

The results of the ROC curve, as seen in Figure 10, show that the AUC is 0.761. An AUC of 0.8 is considered a good value and indicates strong model performance. However, even though the AUC is strong, that does not mean the model is performing well, as seen earlier.

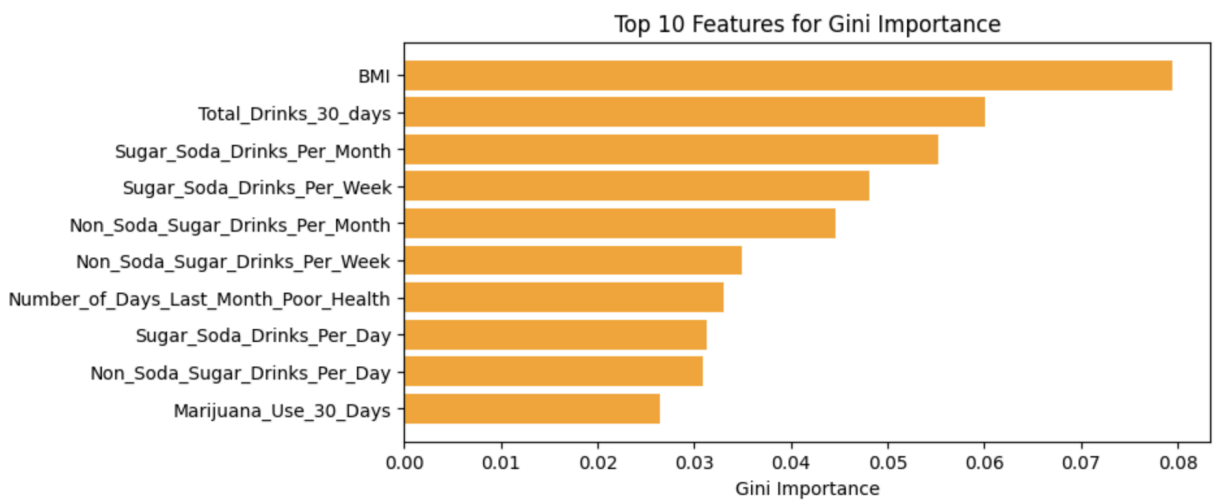

Figure 11: Top 10 Features for Gini Importance

Lastly, feature importance was calculated. The ten features with the highest Gini Importance were included in the results. The table of values from Figure 11 is located in the Appendix. The feature with the highest importance is BMI, indicating that this variable is important for the accuracy of the model. Six of these ten features are part of the sugar drink variable, indicating that diet choices could have an impact on the model. However, this more likely indicates that there is collinearity between these variables, which will need to be addressed.

Discussion & Next Steps

The dataset has a large class imbalance, with 332,416 records representing current non-users of e-cigarettes and only 20,266 records representing current e-cigarette users. This heavy class imbalance is negatively impacting our model performance, leading to inaccurately high accuracy. Therefore, it is not effectively answering our proposed question. In order to improve our models and answer the question, the class imbalance will need to be handled through methods such as class weights or undersampling.

Some key takeaways from the baseline Elastic Net Logistic Regression (and also the Ridge and Lasso models, since the results were the same) was that the model can make a positive prediction correctly but struggles when making a negative class prediction. So, the model can predict whether an individual is an E-Cigarette user but does not do a good job of predicting whether an individual is not an E-Cigarette user. The AUC-ROC score and curve are decent, but the values for precision, recall, and F1-Score are poor. The Ridge and Lasso models also show no different results. Going forward, more iterations will be added to the model to potentially allow the model to converge on better results. The Lasso and Ridge will also be used to see

which, if any, of the features can be excluded or have more bias added to them to potentially increase the model's ability to make predictions.

Key takeaways from the baseline random forest model include poor precision, recall, and F1-score, but a relatively good ROC score. However, considering the overall results and the class imbalance, we are skeptical of these results. It will be interesting to see how the results change with a more balanced dataset and hyperparameter tuning,

For the random forest model, scikit-learn has an attribute called **class_weight** that can address the class imbalance. This was not implemented in the baseline model, so the next step will be testing it by setting class_weight equal to "balanced" to see if model results improve. Additional hyperparameter tuning will include testing a different number of trees (n_estimators) and adjusting the maximum depth of the tree (max_depth) to improve model performance and control complexity. If adjusting class_weight on its own is not sufficient, undersampling methods can also be explored.

The metrics described in the Analysis Plan were all run on the baseline models. The feature_importances_ attribute for the random forest model was also implemented. The Analysis Plan will be adjusted to address multicollinearity among variables in order to improve model performance.

GitHub Link: https://github.com/KWilson310/E-Cigarette-and-Healthy-Habits-Study

Works Cited

"F1_score." *Scikit Learn*, scikit-learn.org/stable/modules/generated/sklearn.metrics.f1_score.html. Accessed 21 Apr. 2026.

Appendix

Data Dictionary:
https://github.com/KWilson310/E-Cigarette-and-Healthy-Habits-Study/blob/main/Reports/Capstone%20Data%20Dictionary%20-%20Pre-Processing%20and%20Feature%20Engineering%20Report.pdf

Random Forest Baseline Feature Importance Table:

Feature	Gini Importance
BMI	0.0794378
Total_Drinks_30_days	0.0600568
Sugar_Soda_Drinks_Per_Month	0.0552671
Sugar_Soda_Drinks_Per_Week	0.0480849
Non_Soda_Sugar_Drinks_Per_Month	0.0446194
Non_Soda_Sugar_Drinks_Per_Week	0.0349259
Number_of_Days_Last_Month_Poor_Health	0.0330663
Sugar_Soda_Drinks_Per_Day	0.0312608
Non_Soda_Sugar_Drinks_Per_Day	0.0309397
Marijuana_Use_30_Days	0.0265024

Random Forest model cross-validation metrics for Recall:

Evaluation Metric	Value
Cross-Validation Recall:	[0.0005 0.001 0.0005 0.0012 0.001]
Mean Cross-Validation Recall	0.0008
Standard Deviation Cross-Validation Recall:	0.0003